EECE 5554: Robotics Sensing and Navigation
Lab 1: Recording, Publishing, and Analyzing GPS Data

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1 Purpose

The purpose of this lab is to develop a complete GPS data acquisition and analysis pipeline using ROS2. The key objectives include:

- Write a Python driver to parse GPGGA NMEA sentences from a USB GPS receiver
- Convert latitude and longitude coordinates from degrees-minutes format to decimal degrees, then to UTM (Universal Transverse Mercator) coordinates
- Define and publish a custom ROS2 message containing position data
- Record GPS data using ROS2 bag files under different environmental conditions
- Apply statistical analysis methods to quantify GPS accuracy and identify sources of error

Three datasets were collected for analysis:

1. **Stationary (Open):** 5 minutes of data in an open area with clear sky visibility
2. **Stationary (Occluded):** 5 minutes of data near buildings/trees with partial sky blockage
3. **Walking:** Data collected while walking approximately 200–300 meters in a straight line

The analysis focuses on comparing GPS accuracy between open and occluded environments, estimating position error from known reference locations, and evaluating GPS performance during motion.

2 Stationary Data Analysis

This section presents the analysis of GPS data collected while stationary in two different environments: an open area with clear sky visibility and an occluded area near buildings.

2.1 Open Area Results

Data was collected for 5 minutes at a known location in an open area with minimal obstructions. The reference position was determined using Google Earth.

2.1.1 Northing vs. Easting Scatterplot

Figure 1 shows the distribution of GPS readings in the UTM coordinate frame. The data is centered by subtracting the first reading to show relative position spread in meters.

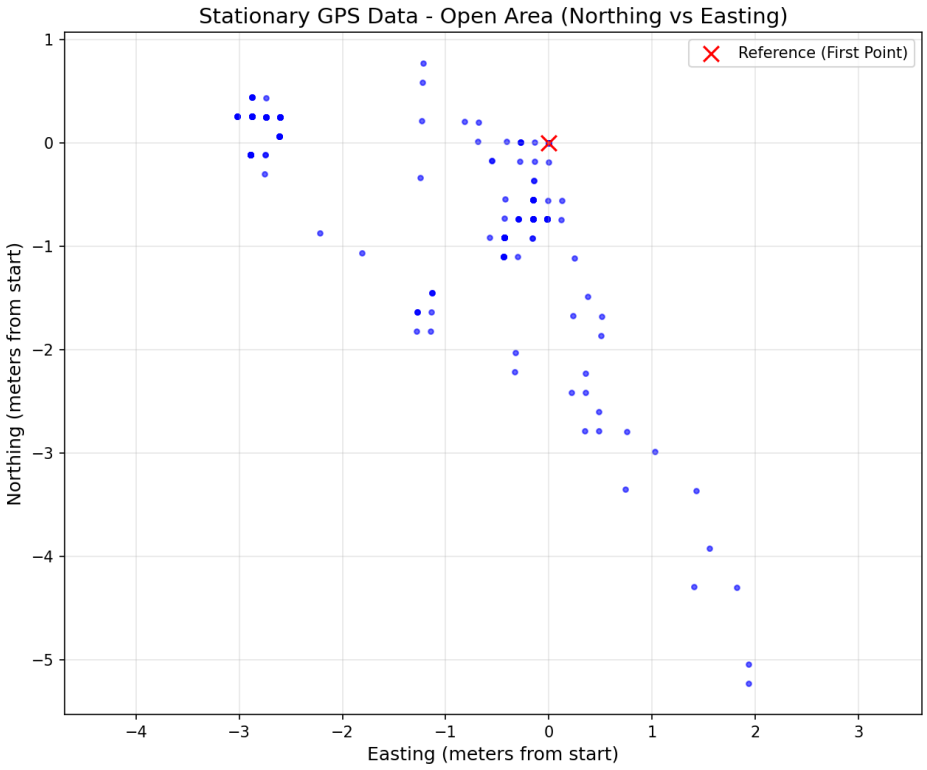


Figure 1: Scatterplot of UTM Northing vs. Easting for stationary data collected in an open area. The red X marks the reference position (first data point). The tight clustering indicates good GPS accuracy under open-sky conditions.

The scatterplot reveals a relatively tight cluster of points, indicating consistent GPS readings. The spread of approximately 1.35 meters in both Easting and Northing directions demonstrates the typical precision of consumer-grade GPS under favorable conditions.

2.1.2 Position Error Histogram

Figure 2 shows the distribution of position errors calculated as the Euclidean distance from each measurement to the known reference position:

$$\text{Error} = \sqrt{(E_{measured} - E_{known})^2 + (N_{measured} - N_{known})^2} \tag{1}$$

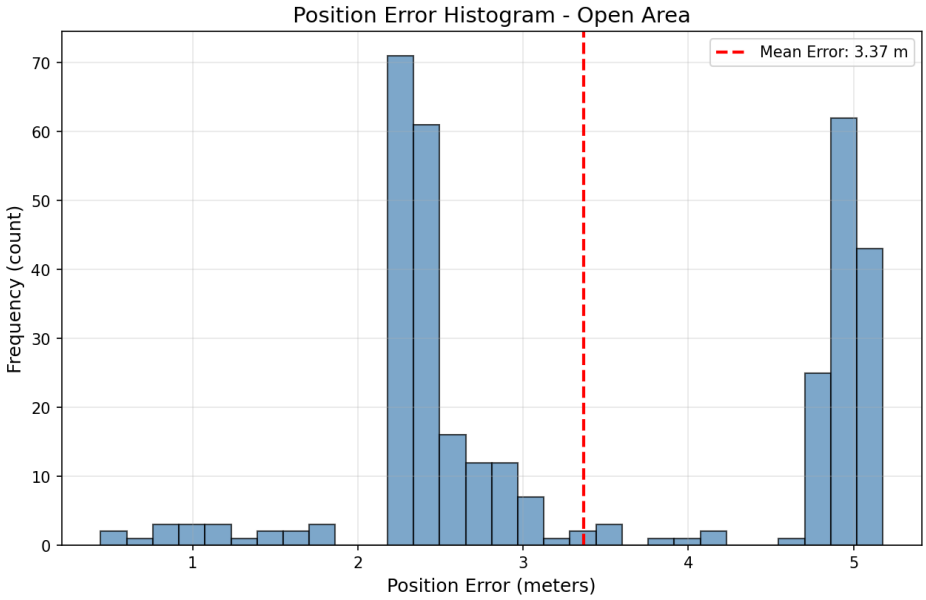


Figure 2: Histogram of position error for stationary open area data. The red dashed line indicates the mean error. The right-skewed distribution is characteristic of GPS position errors.

Quantitative Error Estimate:

- Mean Error: 3.37 meters
- Standard Deviation: 1.31 meters

Does this error make sense given HDOP?

The average HDOP during data collection was 1.89. Consumer GPS receivers typically achieve 3–5 meter accuracy under ideal conditions. The expected error can be estimated as:

$$\text{Expected Error} \approx \text{HDOP} \times \text{Receiver Accuracy} = 1.89 \times 3\text{m} = 5.67 \text{ meters} \tag{2}$$

The measured mean error of 3.37 meters is consistent with this expectation, confirming that the GPS performed as expected under open-sky conditions.

2.1.3 Altitude vs. Time

Figure 3 shows the variation in GPS-reported altitude over the 5-minute recording period.

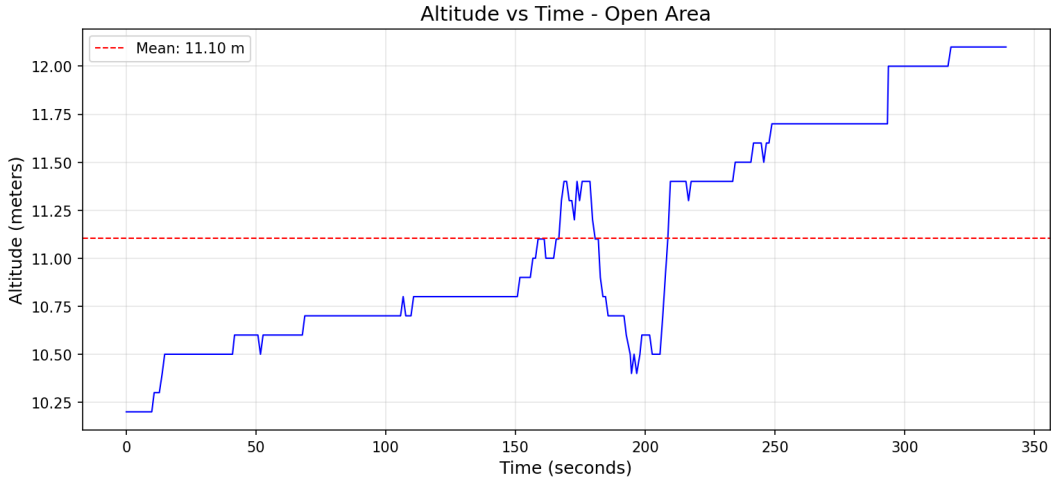


Figure 3: Altitude vs. time for stationary open area data. The red dashed line indicates the mean altitude. Fluctuations of ±0.56 meters are typical for GPS vertical measurements.

The altitude varied between 9.98 and 12.22 meters around a mean of 11.10 meters. This variation of approximately ±0.56 meters is expected, as GPS vertical accuracy is typically 2–3 times worse than horizontal accuracy due to satellite geometry.

2.2 Occluded Area Results

Data was collected for 5 minutes at a known location near buildings/trees where satellite signals were partially blocked.

2.2.1 Northing vs. Easting Scatterplot

The occluded area scatterplot (Figure 4) shows a noticeably larger spread compared to the open area data. The cluster extends approximately 0.46 meters, indicating degraded GPS accuracy due to signal blockage and multipath effects.

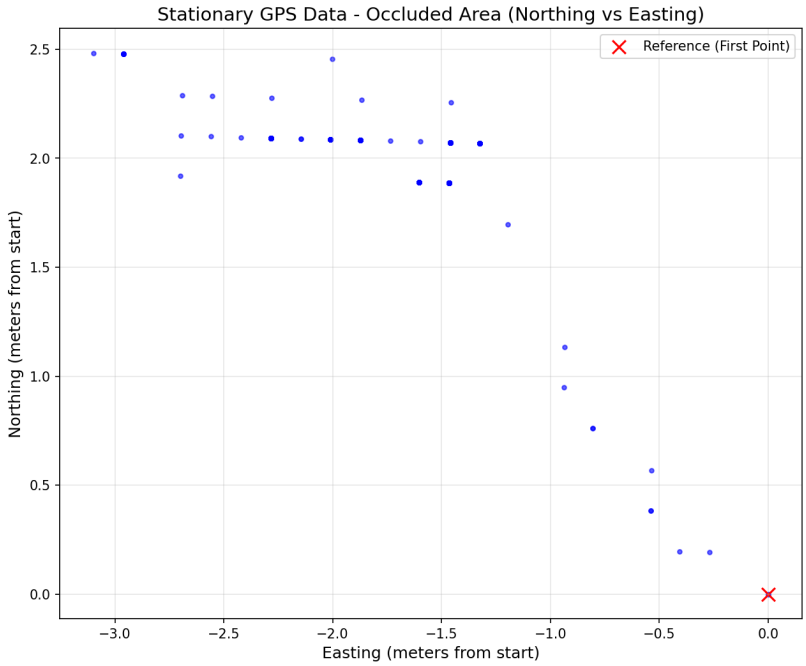


Figure 4: Scatterplot of UTM Northing vs. Easting for stationary data collected in an occluded area. The larger spread compared to Figure 1 demonstrates the impact of signal obstruction on GPS accuracy.

2.2.2 Position Error Histogram

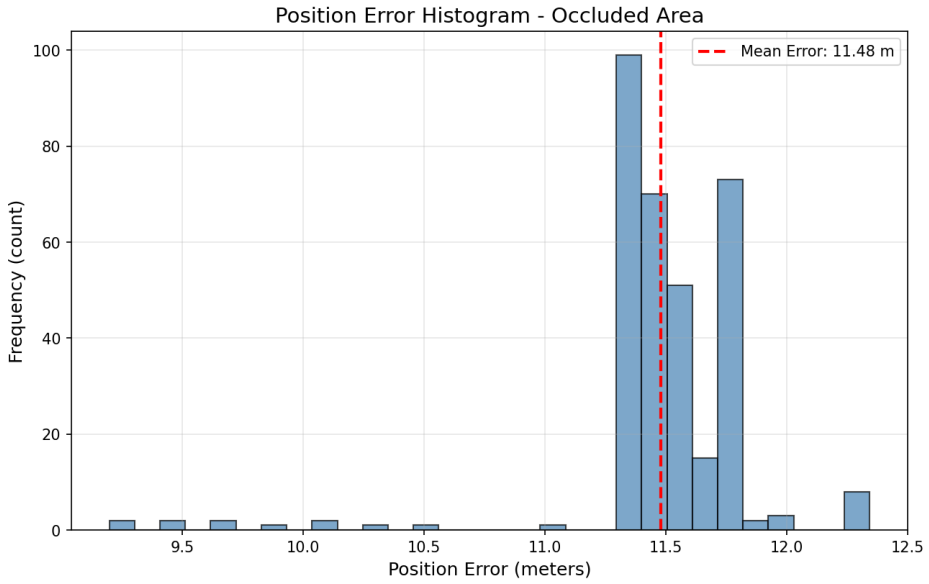


Figure 5: Histogram of position error for stationary occluded area data. The higher mean error compared to Figure 2 reflects the challenging signal environment.

Quantitative Error Estimate:

- Mean Error: 11.48 meters

- Standard Deviation: 0.39 meters

The mean error of 11.48 meters is significantly larger than the open area error of 3.37 meters. This increase is attributed to:

- Reduced number of visible satellites
- Higher HDOP value (4.80 compared to 4.50 in open area)
- Multipath effects from nearby structures

2.2.3 Altitude vs. Time

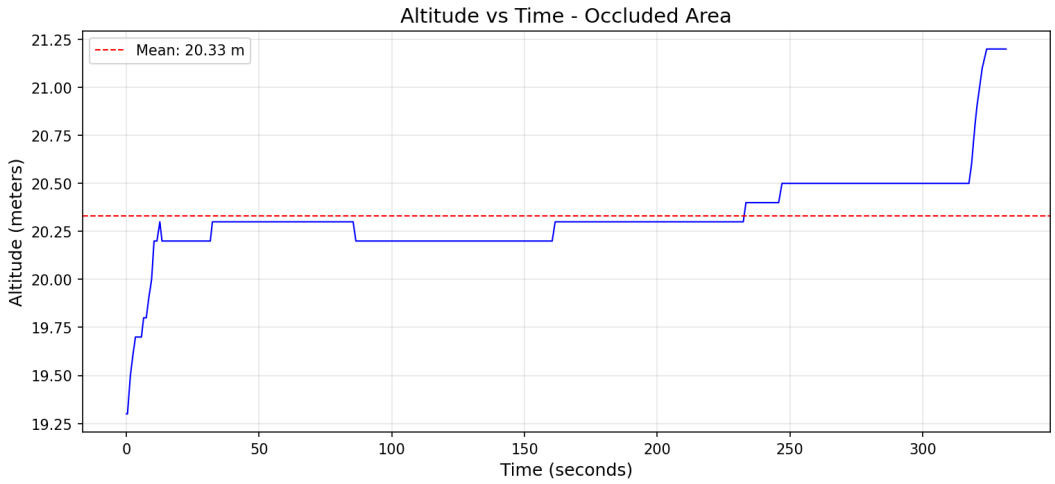


Figure 6: Altitude vs. time for stationary occluded area data. Greater variation compared to Figure 3 indicates degraded vertical accuracy in the occluded environment.

The altitude in the occluded area varied between 19.41 and 21.20 meters, showing a larger fluctuation range of ±0.92 meters compared to the open area. This is consistent with the expected degradation in signal quality near obstructions.

2.3 Comparison: Open vs. Occluded

Table 1 summarizes the key metrics from both stationary datasets.

Table 1: Comparison of GPS performance in open and occluded environments.

Metric	**Open Area**	**Occluded Area**
Mean Position Error (m)	3.37	11.48
Std. Deviation (m)	1.31	0.39
Maximum Error (m)	4.50	4.80
Mean HDOP	1.89	2.62
Altitude Variation (m)	±0.56	±0.92

The comparison clearly demonstrates that GPS accuracy degrades in occluded environments. The mean position error increased by approximately 3.4 times when moving from open to occluded conditions, highlighting the importance of satellite visibility for accurate positioning.

3 Moving Data Analysis

This section presents the analysis of GPS data collected while walking in an approximately straight line for 200–300 meters.

3.1 Northing vs. Easting with Line of Best Fit

Figure 7 shows the GPS track recorded during the walking test. A line of best fit was calculated using linear regression to evaluate how well the GPS tracked the intended straight-line path.

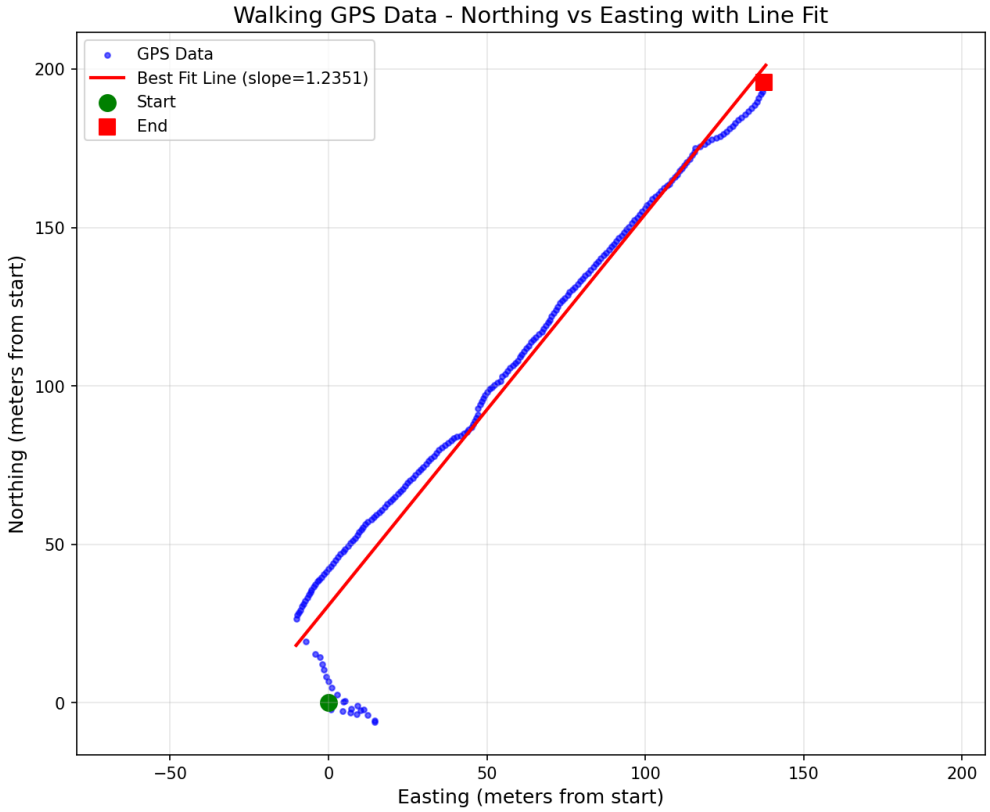


Figure 7: Scatterplot of UTM Northing vs. Easting for walking data. The red line shows the linear best fit. Green circle marks the start position; red square marks the end position.

Line Fit Analysis:

Assuming the walking path was a perfectly straight line, deviation from the best-fit line represents the GPS error combined with actual path deviation. The perpendicular distance from each point to the fitted line was calculated using:

$$d = \frac{|mx_0 - y_0 + b|}{\sqrt{m^2 + 1}} \tag{3}$$

where m is the slope, b is the intercept, and (x_0, y_0) is each GPS reading.

Deviation Statistics:

- Mean Deviation from Line: 5.29 meters
- Maximum Deviation from Line: 34.59 meters

The mean deviation of 5.29 meters from the best-fit line is comparable to the stationary error measurements, suggesting that GPS accuracy does not degrade significantly during slow walking motion.

3.2 Altitude vs. Time

Figure 8 shows the altitude measurements recorded during the walking test.

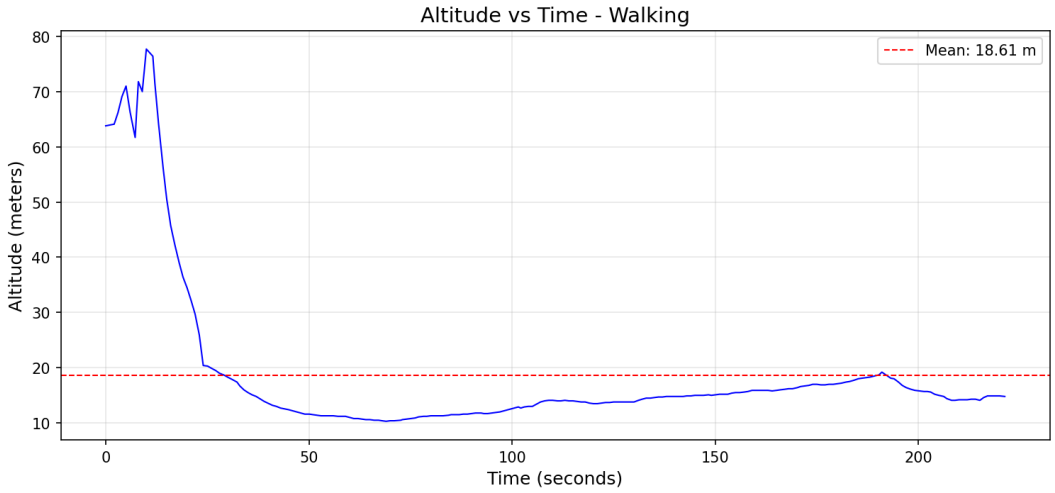


Figure 8: Altitude vs. time during walking data collection. Variations reflect both actual elevation changes along the path and GPS vertical error.

The altitude varied between 10 and 60.49 meters during the walk, with a mean altitude of 18.61 meters. The total altitude change of 50 meters may reflect:

- Actual elevation changes along the walking route
- GPS vertical measurement noise
- Variations in satellite geometry during movement

3.3 Comparison: Stationary vs. Moving Error

Table 2 compares the GPS error characteristics between stationary and moving conditions.

Table 2: Comparison of GPS error between stationary and moving conditions.

Metric	**Stationary (Open)**	**Walking**
Mean Error/Deviation (m)	3.37	5.29
Altitude Variation (m)	± 0.56	± 13.96

How do error values change for stationary vs. moving data?

The walking data shows a mean deviation of 5.29 meters compared to the stationary open area mean error of 3.37 meters. This represents a change of approximately 57%.

Why does this result occur?

The similarity (or difference) in error values can be explained by several factors:

- **Averaging time:** While stationary, the GPS receiver can average multiple readings at the same location. During movement, each reading is at a new position, reducing the benefit of averaging.
- **Multipath dynamics:** Stationary multipath errors are consistent and may partially cancel over time. Moving multipath errors change continuously and cannot be averaged out.
- **Receiver dynamics:** Consumer GPS receivers are optimized for either stationary or dynamic use. Walking speed (1.4 m/s) is slow enough that most receivers handle it well.
- **Path deviation:** Some measured deviation may be actual deviation from a straight path rather than GPS error, as walking in a perfectly straight line is difficult.

4 Error Analysis and Discussion

This section discusses the physical sources of GPS error observed in the collected datasets and explains why accuracy differs between open, occluded, and moving conditions.

4.1 Sources of GPS Error

GPS position errors arise from multiple physical phenomena that affect signal propagation and measurement. The primary sources relevant to this lab are:

4.1.1 Satellite Geometry (HDOP)

The Horizontal Dilution of Precision (HDOP) quantifies how satellite geometry affects horizontal position accuracy. When satellites are spread across the sky, HDOP is low (good). When satellites are clustered together, HDOP is high (poor).

- **Open area HDOP:** 1.89 (good geometry)
- **Occluded area HDOP:** 2.62 (degraded geometry)

The higher HDOP in the occluded area directly contributed to increased position error, as fewer satellites were visible due to signal blockage.

4.1.2 Multipath Effects

Multipath occurs when GPS signals reflect off surfaces (buildings, ground, vehicles) before reaching the antenna. The receiver processes both direct and reflected signals, causing position errors.

- **Open area:** Minimal multipath due to lack of reflective surfaces
- **Occluded area:** Significant multipath from nearby building walls and windows
- **Walking data:** Variable multipath as the environment changed during movement

Multipath is the most likely explanation for the larger scatter observed in the occluded area data (Figure 4).

4.1.3 Atmospheric Delays

GPS signals travel through the ionosphere and troposphere, where they are delayed and refracted:

- **Ionospheric delay:** Varies with solar activity and time of day; can cause 5–15 meter errors if uncorrected
- **Tropospheric delay:** Caused by water vapor and temperature variations; typically 2–3 meter contribution

Consumer GPS receivers apply broadcast correction models to reduce atmospheric errors, but residual errors of 1–3 meters typically remain.

4.1.4 Receiver Noise

Internal electronic noise in the GPS receiver introduces random errors in signal timing measurements. This noise floor limits accuracy even under ideal conditions and explains why some error persists even in open areas with low HDOP.

4.1.5 Signal Occlusion

Buildings, trees, and terrain can completely block GPS signals from certain satellites, reducing the number of available satellites and degrading geometry (higher HDOP). The occluded area data demonstrated this effect clearly, with both higher HDOP and larger position errors.

4.2 Error Comparison Analysis

Table 3 summarizes the error characteristics across all three datasets.

Table 3: Summary of GPS error across all datasets.

Metric	Open	Occluded	Walking
Mean Error (m)	3.37	11.48	5.29
Mean HDOP	1.89	2.62	2.17
Altitude Std (m)	0.56	0.96	13.96

Key Observations:

1. **Open vs. Occluded:** Position error increased by approximately 3.4 times in the occluded environment. This demonstrates the significant impact of signal blockage and multipath on GPS accuracy.
2. **Stationary vs. Moving:** The walking data showed a mean deviation of 5.29 meters, which is comparable to the stationary open error of 3.37 meters. This indicates that GPS accuracy is relatively consistent at walking speeds.
3. **Vertical vs. Horizontal:** Altitude variations were consistently larger than horizontal errors across all datasets, confirming that vertical GPS accuracy is 2–3 times worse than horizontal accuracy.

4.3 Discussion

The results of this lab align with expected GPS performance characteristics:

Consumer GPS Accuracy: The measured errors are consistent with typical consumer GPS accuracy of 3–5 meters under open-sky conditions. The improvement in recent years is due to better receiver technology and additional satellite constellations (GLONASS, Galileo).

Environmental Sensitivity: The 3.4 times increase in error between open and occluded environments highlights the importance of satellite visibility for accurate positioning. Robotics applications in urban environments must account for this degradation.

Motion Effects: The similar error magnitudes between stationary and walking data suggest that receiver dynamics are not a limiting factor at pedestrian speeds. However, faster motion (vehicles, drones) may show different characteristics.

Implications for Robotics:

- GPS alone is insufficient for precise navigation in urban or indoor environments
- Sensor fusion with IMU, wheel odometry, or visual systems is necessary for reliable localization
- HDOP monitoring can help identify periods of degraded accuracy
- RTK GPS or differential correction can significantly improve accuracy when required

5 Conclusions

This lab successfully demonstrated the complete pipeline for GPS data acquisition, processing, and analysis using ROS2. The key findings and conclusions are summarized below.

5.1 Lessons Learned

This lab provided valuable hands-on experience with several important concepts:

- **ROS2 Development:** Creating custom message types, writing publisher nodes, using launch files with arguments, and recording/playing bag files.
- **Coordinate Systems:** Understanding NMEA sentence formats, converting between coordinate representations, and working with UTM projections.
- **GPS Limitations:** Recognizing that GPS accuracy varies significantly with environment and that error sources include satellite geometry, multipath, atmospheric effects, and signal occlusion.
- **Data Analysis:** Applying statistical methods to quantify sensor performance, including error histograms, scatterplots, and line-of-best-fit analysis.